

This document contains linear-algebra questions covering background material for the APPM PhD program at CU-Boulder. For each question simply indicate how confident you are of being able to answer it correctly. All vectors are lists of real numbers unless specified otherwise.

1. Can you find, by hand, an expression that describes all solutions of a system of linear equations, whenever at least one solution exists?
2. Can you find the rank of a $m \times n$ matrix by hand?
3. Suppose that $\mathbf{A}$ is an $m \times n$ matrix with rank n and $\mathbf{1}$ is an $n \times n$ matrix of ones. Prove that

$$\mathbf{A} \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \right)$$

has rank exactly $n - 1$.

4. Suppose that you are given $\mathbf{A}$, $\mathbf{B}$, and $\vec{x}$. Do you know how to evaluate the expression $(\mathbf{A} + \mathbf{B}^{-1})^{-1} \vec{x}$ without computing the inverse of any matrices?
5. Can you find a basis for the range (aka image, aka column space), the co-range (aka co-image, aka row space), the kernel (aka null space), and the co-kernel (aka left hand null space) of an $n \times m$ matrix by hand?
6. Suppose that you are given P pairs of vectors $(\vec{x}_1, \vec{b}_1), \dots, (\vec{x}_P, \vec{b}_P)$ all of which satisfy

$$\mathbf{A} \vec{x}_p = \vec{b}_p, \quad p = 1, \dots, P$$

where $\mathbf{A}$ is an $m \times n$ matrix that is not given to you. If you are given a vector $\vec{b}$ that is in the span of $\{\vec{b}_1, \dots, \vec{b}_P\}$, can you find a solution $\vec{x}$ to $\mathbf{A} \vec{x} = \vec{b}$? (The question is not asking whether a solution exists, which it does. It's asking if you know how to find it without knowing the matrix $\mathbf{A}$.)

7. Consider the linear operator $\mathcal{L} : f \mapsto xf'$ acting on the space of polynomials of degree ≤ 5 . Can you find a matrix representation of this operator with respect to the monomial basis?
8. Consider the linear, non-autonomous system of ordinary differential equations

$$\frac{d\vec{x}}{dt} = \mathbf{L}(t)\vec{x}.$$

Prove that there is a matrix $\mathbf{\Pi}(t)$ such that $\mathbf{\Pi}(t)\vec{x}_0$ is the solution to the differential equation at time t starting from initial condition $\vec{x}(t=0) = \vec{x}_0$.

9. Do you know the definitions of the p norms of vectors and matrices for $1 \leq p \leq \infty$?
10. Suppose that $\mathbf{A}$ is an $m \times n$ matrix. Can you prove that $\mathbf{A}^T \mathbf{A}$ is non-negative definite? Could you state necessary and sufficient conditions on $\mathbf{A}$ so that $\mathbf{A}^T \mathbf{A}$ is positive definite?
11. Define $f : (\vec{x}, \vec{y}) \mapsto \vec{x}^T \mathbf{K} \vec{y}$ for a square matrix $\mathbf{K}$. Under what conditions on $\mathbf{K}$ does $f(\vec{x}, \vec{y})$ define an inner product on $\mathbb{R}^n$?
12. Give an expression for the adjoint of $\mathbf{A}$ with respect to the inner product defined by $\vec{x}^T \mathbf{K} \vec{y}$.
13. Given two vectors in $\mathbb{R}^n$, can you find the Euclidean angle between them?

14. Given an $m \times n$ matrix $\mathbf{A}$ and a vector $\vec{b}$ in $\mathbb{R}^m$, can you find the orthogonal projection of $\vec{b}$ onto the range of $\mathbf{A}$? (Assume orthogonality is defined with respect to the dot product.)
15. Given a nonzero $m \times n$ matrix $\mathbf{A}$, can you find an *orthonormal* basis for its range? (The question is not asking whether such a basis exists; it is asking whether you know how to find one.)
16. Can you prove the following statement: If $\mathbf{A}$ is skew-symmetric, then

$$(\mathbf{I} - \mathbf{A})(\mathbf{I} + \mathbf{A})^{-1}$$

is well-defined and is an orthogonal matrix with determinant equal to 1?

17. Prove that the eigenvalues of $\mathbf{A}$ are also eigenvalues of $\mathbf{A}^T$ without making reference to the properties of the determinant.
18. Suppose that there is an invertible matrix $\mathbf{S}$ such that $\mathbf{SA} = \mathbf{BS}$. Prove that $\mathbf{A}$ and $\mathbf{B}$ have the same eigenvalues.
19. Give an example of a 2×2 matrix that is not diagonalizable.
20. Suppose that $\mathbf{A}$ is symmetric and that $\mathbf{A} = \mathbf{LU}$ is an LU factorization of $\mathbf{A}$. If the diagonal values of $\mathbf{U}$ are all positive, what can you conclude, if anything, about the eigenvalues of $\mathbf{A}$?
21. Prove that $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$ for every $\mathbf{A} \in \mathbb{R}^{n \times n}$ and every operator norm $\|\cdot\|$. $\rho(\mathbf{A})$ is the spectral radius of $\mathbf{A}$.
22. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have full column rank. Prove that $\|\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}\|_2 = 1/\sigma_n$ where σ_n is the n^{th} singular value of $\mathbf{A}$. (Note that $\mathbf{A}^{-1}$ is not defined, so your proof cannot use $\mathbf{A}^{-1}$.)
23. Prove the following: the iteration defined by $\vec{x}_{k+1} = \mathbf{B}\vec{x}_k$, where $\mathbf{B} \in \mathbb{R}^{n \times n}$ and $\vec{x}_k \in \mathbb{R}^n$, converges to $\vec{0}$ for every initial condition $\vec{x}_0$ if and only if $\rho(\mathbf{B}) < 1$ where $\rho(\mathbf{B})$ is the spectral radius of $\mathbf{B}$.
24. Prove that if $\mathbf{A}$ is skew symmetric, then $e^{\mathbf{A}}$ is orthogonal.
25. Prove the following: If $\|\mathbf{F}\| < 1$ where $\|\cdot\|$ is an operator norm, then

$$(\mathbf{I} - \mathbf{F})^{-1} - \mathbf{I} = \mathbf{F}(\mathbf{I} - \mathbf{F})^{-1}.$$

Computing questions:

1. Can you write up homework solutions in $\text{\LaTeX}$ and print them out?
2. Can you use `pip` or `conda` (or `mamba`) to set up and manage Python environments?
3. Can you use a Jupyter notebook?
4. Can you use `numpy` or `scipy` to perform numerical computations? (Or Matlab or Julia)
5. Can you plot a mathematical function (e.g. using `matplotlib.pyplot`), save it as a file, include it in a $\text{\LaTeX}$ document, and print it out as pdf?